Investment Risk & Probability Modeling

Probabilistic Modeling and Monte Carlo Simulation

Alaska Airlines

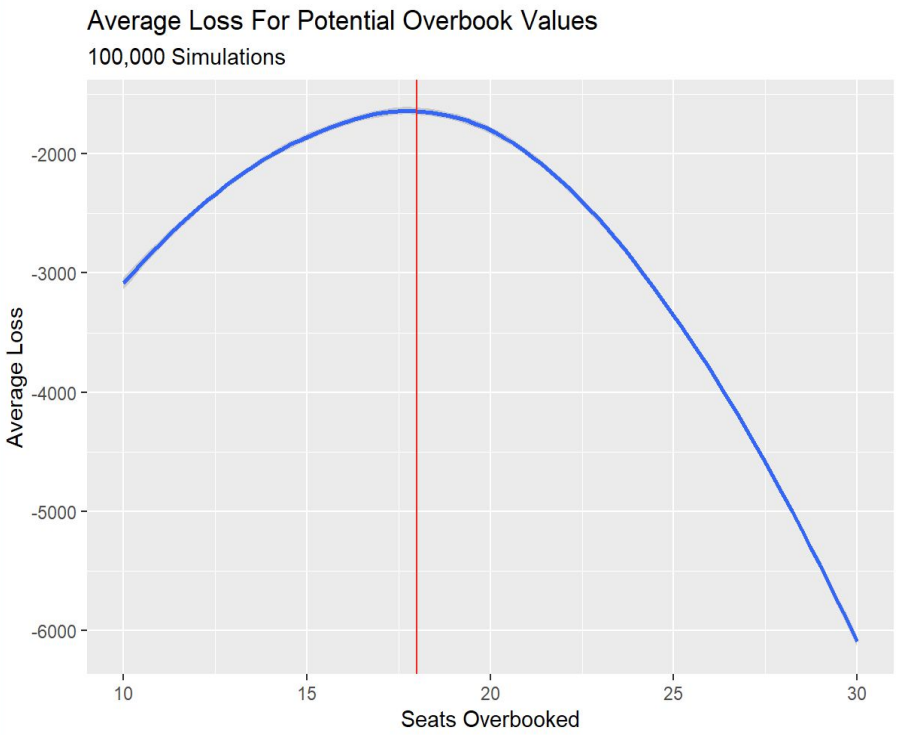
Business Problem:

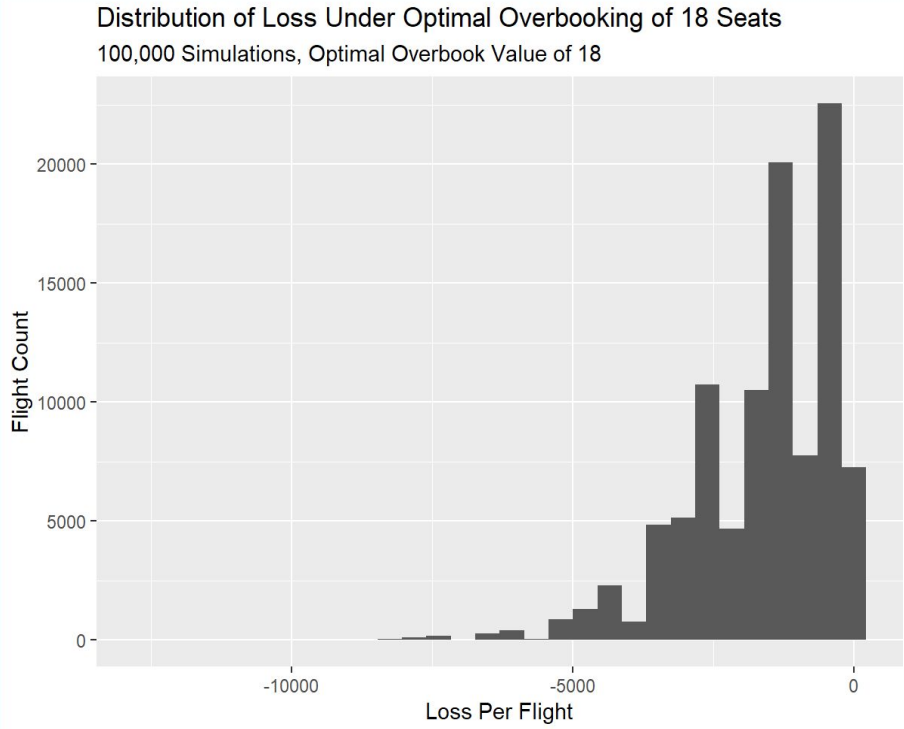
Almost every Alaska Airlines flight has a certain number of passengers that do not show up for the flight referred to as "no-shows." Airlines will typically overbook flights to account for these no-shows, but Alaska Airlines needs to determine exactly how many seats it should overbook per flight. While the cost of an unused seat is $300, the cost of a voucher for a bumped passenger is double at $600. Alaska airlines believes that the number of no-shows per flight is normally distributed with a mean of 20 and a standard deviation of 5.

Recommendation:

Overbook 18 seats per flight because it minimizes the average loss to around $1630 per flight.

This falls to the left of the average of no-shows, but this makes sense considering that addressing a bump in passengers is twice as expensive as a no-show.





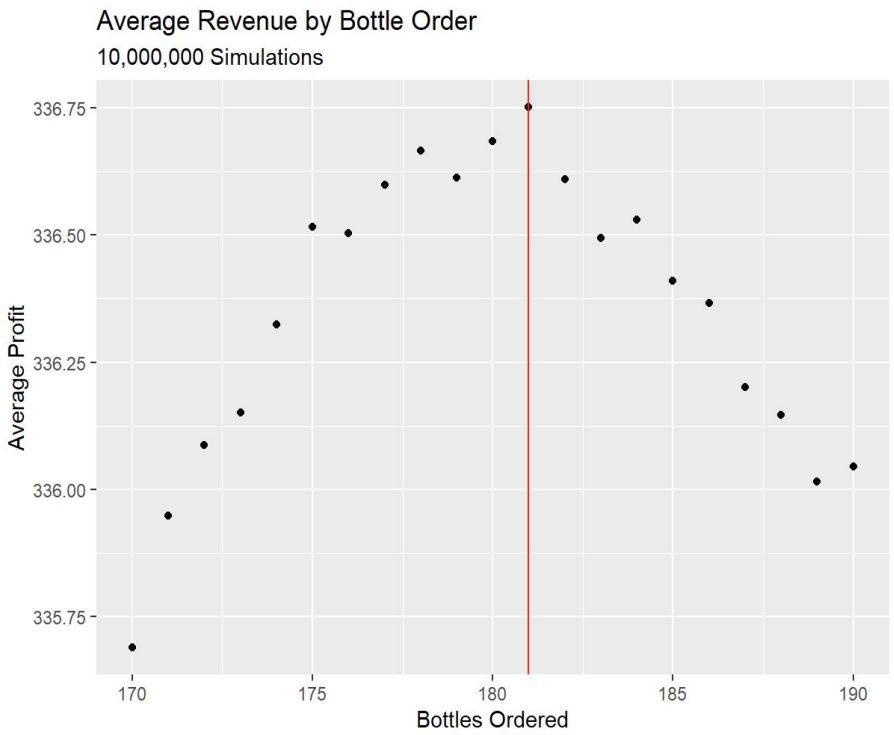
Bronco Wine

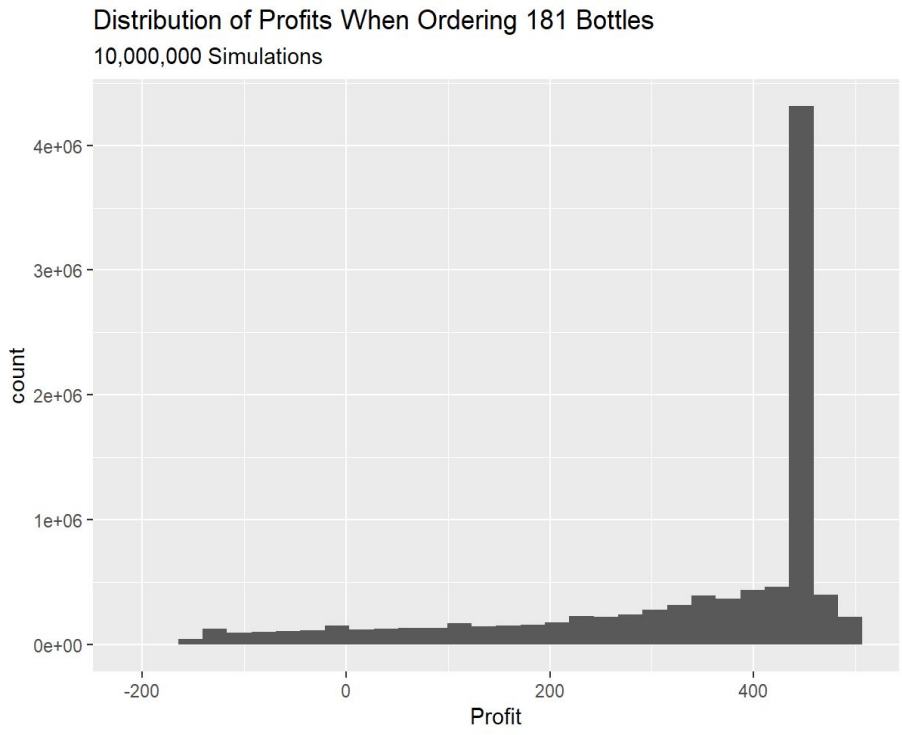
Business Problem:

Each year Bronco Wine must decide how many bottles to order. They believe they can sell between 100 and 300 bottles of wine with demand following a uniform distribution. They sell their bottles for $10 and each bottle typically costs $7.50, but if the supplier is unable to meet their order, they cost $7.25 instead. Bottle supply follows a normal distribution with mean of 200 and standard deviation of 5. Unsold bottles can be returned for $2.50. Bronco must decide the optimal number of bottles to order to maximize profit.

Recommendation:

Order 181 bottles because it maximized profit to around $337. It should be noted that results past around 179 to around 210 bottles ordered are exceedingly similar even under 10,000,000 simulations. 181 happened to be the optimal value under this simulation, but if orders are more easily processed in more standard increments, 200 bottles ordered would also produce a near-optimal value.



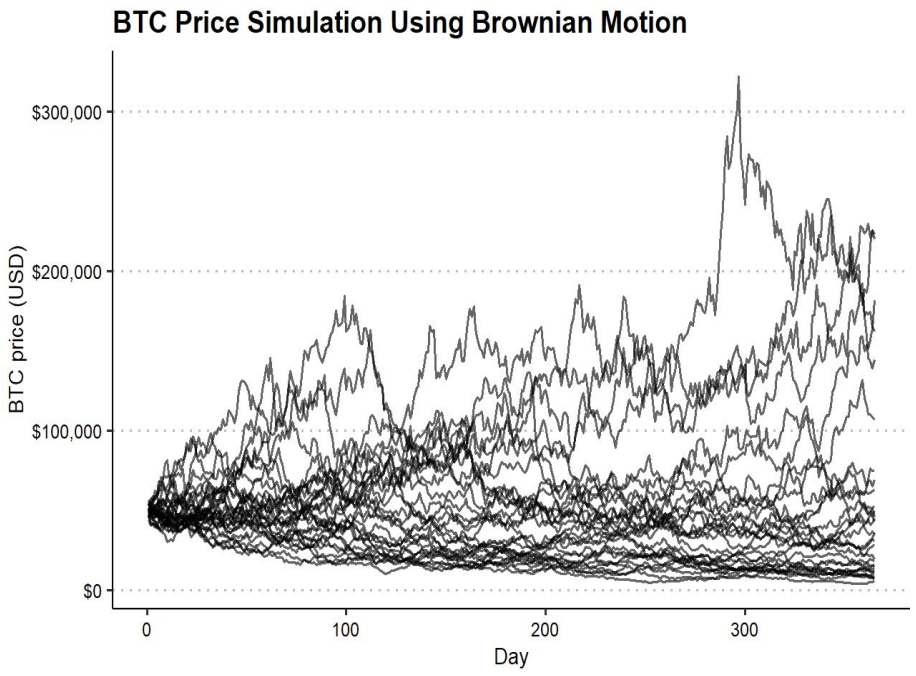
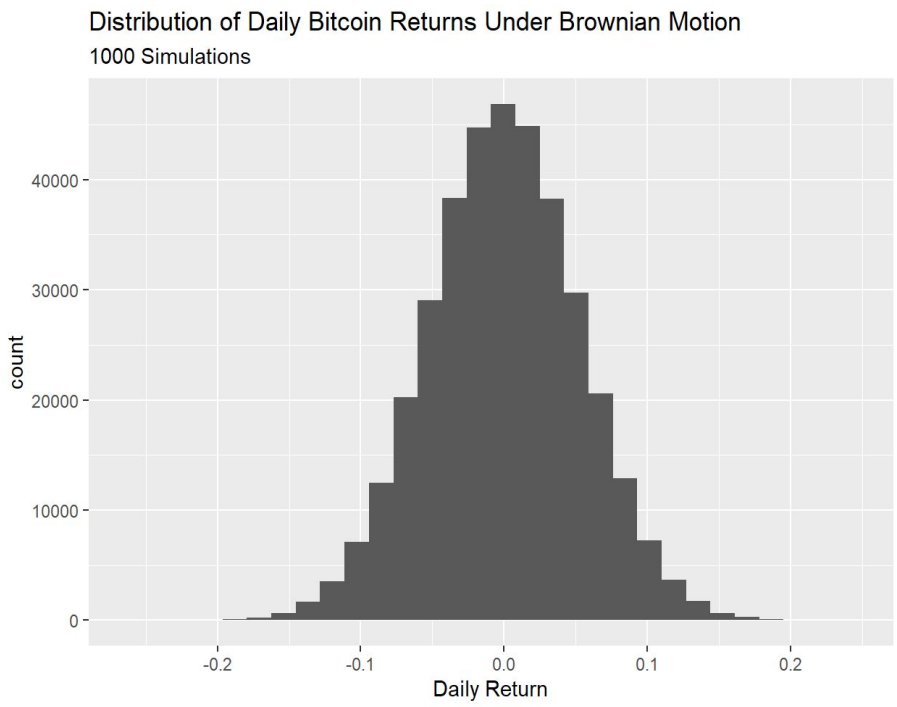


Bitcoin Price

Business Problem:

Bitcoin prices fluctuate with a mean annual return of 40% and a standard deviation of 100%. The task at hand is to model Bitcoin price fluctuations using Brownian motion to see where the price lies after one year assuming prices start at $50,000.

Mean	Bear (Q1)	Base (Q2)	Bull (Q3)
$48,293	$13,643	$27,306	$58,680

Recommendation:

The output of the Brownian motion simulation shows that the final price in the base case is approximately $27,300 which is almost half of what the starting price was. The bull case is improved somewhat with a final value of approximately $58,000, but that is not enough to justify investment. If the question at hand is whether to invest in Bitcoin, this simulation shows that it is not a sound investment.